IBVAP - Project Guide

Intelligent Border Video Analytics Platform | Problem Statement ID: 26187

1. Project Overview

Ek **software platform** banana hai jo border pe pehle se lage normal CCTV cameras ko **smart cameras** bana de, bina koi naya hardware lagaye. Software camera ki live feed (RTSP stream) lega, uspe AI chalayega, aur kuch suspicious dikhte hi khud alert bhejega.

Formula: Purana CCTV camera + AI software = Intelligent surveillance system

One-liner (overview maange to)

> *"IBVAP ek AI software hai jo border pe lage existing CCTV cameras ko bina naya hardware lagaye intelligent surveillance system mein badal deta hai."*

30-second pitch

> *"Border pe hazaaron CCTV cameras lage hain, par wo sirf recording karte hain. Unhe monitor karne ke liye jawan ko lagatar screen dekhni padti hai. FRS aur ANPR jaise features ke liye mehenga specialized hardware chahiye. Hamara platform IBVAP existing IP cameras ki live feed leke AI aur computer vision se real-time mein insaan aur vehicle detect karta hai, unhe track karta hai, number plate padhta hai, chehre pehchanta hai, aur virtual fence cross hone ya suspicious activity pe turant alert bhejta hai. Ye din aur raat dono mein kaam karta hai, edge pe offline chal sakta hai, aur existing command & control systems se integrate ho jata hai."*

Key USPs: No new hardware | Real-time alerts | Edge/offline capable | Low cost | Scalable | Human workload kam

Architecture flow

CCTV cameras -> Video ingestion (RTSP) -> AI engine (detection, tracking, face, ANPR, rules) -> Alert engine -> Dashboard + Database + API

2. Important Terms

Term	Full form	Matlab (in short)
IBVAP	Intelligent Border Video Analytics Platform	Aapke project ka naam. Border CCTV videos ko AI se analyze karne wala platform.
ANPR	Automatic Number Plate Recognition	Vehicle ki number plate AI se khud padhna aur blacklist/database se check karna.
FRS	Facial Recognition System	Chehra dhoondhna (detection) aur pehchanna ki wo kiska hai (recognition), jaise watchlist match.
CCTV	Closed-Circuit Television	Normal security camera.
IP Camera	Internet Protocol Camera	Network se juda camera jiski live feed software mein le sakte hain.
RTSP	Real Time Streaming Protocol	Wo tareeka jisse IP camera live video bhejta hai.
BOP	Border Out Post	Border pe security forces ki chowki.
Virtual fence	-	Video mein draw ki gayi nakli line/zone. Koi cross kare to alert.
C2 system	Command & Control	Central control room jahan se saari monitoring aur decisions hote hain.

Term	Full form	Matlab (in short)
Edge deployment	-	AI ko cloud ki jagah local device pe chalana, bina internet ke.

3. 1 Week MVP Scope

Pakka banao (Core)

- Video input: recorded video / webcam / phone ko live feed ki tarah chalana
- Human + vehicle detection with tracking
- Virtual fence intrusion: line/zone cross hote hi alert
- ANPR: plate detect + number padhna
- Dashboard: live video, alerts with snapshot, event log

Time bache to (Bonus)

- Face detection + watchlist matching
- Loitering detection (ek jagah zyada der rukna)
- Night video demo (low-light enhancement ke saath)

Skip karo (sirf 'Future scope' mein bolo)

- Complex suspicious activity (crawling, fighting)
- Real edge device deployment (Jetson etc.)
- Real C2 system integration
- Apna model scratch se train karna

4. Free Software & Tools (Total cost: Rs. 0)

Tool	Kaam	Install / Link	Imp notes
Python 3.10+	Main programming language	python.org	Sab kuch isi mein banega.
VS Code	Code editor	code.visualstudio.com	Python extension install kar lena.
Google Colab	Free GPU pe code chalana	colab.research.google.com	Laptop mein GPU nahi hai to ye use karo. Session time-limited hota hai.
Kaggle Notebooks	Free GPU (backup)	kaggle.com	Approx 30 ghante/week GPU free.
Ultralytics YOLO (v8 / 11)	Human & vehicle detection	pip install ultralytics	Pretrained aata hai. Slow laptop pe 'n' (nano) model use karo.
supervision (Roboflow)	Tracking (ByteTrack), zones, line crossing, drawing	pip install supervision	Virtual fence aur tracking ka sabse bada shortcut. 2-3 din bachayega.
OpenCV	Video read/write, frames, image processing	pip install opencv-python	RTSP stream bhi isi se read hota hai.
EasyOCR	Number plate ka text padhna (ANPR)	pip install easyocr	Setup aasaan. Pehli baar model download hota hai.
PaddleOCR	OCR (EasyOCR ka alternative)	pip install paddleocr	Accuracy aksar better, setup thoda heavy.
InsightFace	Face detection + recognition	pip install insightface	High accuracy. Pretrained models sirf non-commercial use ke liye, hackathon ke liye theek.
face_recognition	Simple face recognition	pip install face_recognition	Aasaan API, par dlib install Windows pe mushkil ho sakta hai.
FastAPI	Backend / REST API	pip install fastapi uvicorn	C2 integration ke liye API yahin se dikhao.
SQLite	Database (alerts, event log)	Python ke saath built-in	Koi server setup nahi chahiye.
Streamlit	Dashboard (sabse fast)	pip install streamlit	Pure Python mein UI. React aati ho to wo bhi chalega.
Roboflow (free tier)	Datasets, pretrained models, labeling	roboflow.com / universe.roboflow.com	Indian number plate model yahin milega.
LabelImg / CVAT	Data labeling	github.com/HumanSignal/labelImg, cvat.ai	Sirf tab jab khud model train karna pade.
GitHub	Code share + version control	github.com	Dono log ek repo mein kaam karo.

Live CCTV ka jugaad (bina camera kharide)

Tool	Kaam	Imp notes
IP Webcam (Android app)	Phone ko IP camera bana deta hai	Stream URL deta hai jo asli CCTV ki tarah software mein chalta hai. Phone aur laptop same WiFi pe hone chahiye.
MediaMTX	Recorded video ko RTSP stream banana	Demo mein recorded video bhi 'live camera' jaisi lagegi. FFmpeg ke saath use karo.
FFmpeg	Video convert / stream push karna	MediaMTX ko video bhejne ke liye.

Tool	Kaam	Imp notes
VLC Player	Stream test karna	Media -> Open Network Stream mein RTSP URL daal ke check karo.

5. Data Sources (Testing & Demo ke liye)

Pretrained models use kar rahe ho, isliye data sirf **testing aur demo** ke liye chahiye, training ke liye nahi.

Category	Source	Kya milega / Note
Demo videos	Khud phone se record karna (BEST)	College gate, parking, road, raat ka movement. No copyright issue.
Demo videos	Pexels.com, Pixabay.com	Free stock videos. Search: CCTV, traffic, people walking, night street.
Surveillance	VIRAT dataset	Asli surveillance footage, humans + vehicles.
Suspicious activity	UCF-Crime dataset (Kaggle)	CCTV clips of anomalous / criminal activity.
Number plates	Roboflow Universe	Search 'Indian number plate'. Dataset + pretrained model dono.
Number plates	Kaggle	Search 'Indian vehicle number plate dataset'.
Number plates	Khud ki photos	Parking mein 15-20 gaadiyon ki photo/video.
Faces (watchlist)	Apni + dosto ki photos	Permission leke. Demo mein 'wanted person' ki tarah pehchanwao.
Faces (testing)	LFW (Labeled Faces in the Wild)	Standard face recognition test dataset.
Night / low-light	ExDark dataset	Low-light images.
Night / infrared	LLVIP dataset	Infrared + visible camera paired images.

Note: YouTube videos demo mein use karne se bacho (copyright). Pexels/Pixabay aur khud ki recording safe hai.

6. 7-Day Plan (2 Members)

Person A (AI side): detection, tracking, fence, ANPR, face
Person B (App side): backend, database, dashboard, alerts, PPT, demo video

Din	Person A (AI)	Person B (App)
Day 1	Setup, YOLO se human/vehicle detection chalana	Test videos jama karna, dashboard ka layout
Day 2	Tracking + virtual fence intrusion	FastAPI backend + SQLite, alert save karna
Day 3	ANPR (plate detection + OCR)	Dashboard pe live feed + alerts dikhana
Day 4	Face detection + watchlist	AI aur dashboard ko jodna (integration)
Day 5	Loitering / night demo (bonus)	Alert snapshot, sound, event log, filters
Day 6	Dono milke: bug fixing, testing, sab ek saath chalana	
Day 7	Dono milke: demo video record, PPT finalize, submission	

7. Zaroori Tips

- Submission mein exactly kya maanga hai (PPT / prototype / video) pehle confirm kar lo.
- **Day 4 tak integration shuru kar do.** Aakhri din jodne mein sab toot jata hai.
- Demo ke liye clear daylight videos chuno, wo sabse impressive lagte hain.
- GPU nahi hai to yolov8n / yolo11n (nano) model use karo ya Colab pe chalao.
- **Demo video pehle se record kar lo.** Live demo fail ho to backup rahega.
- PPT mein architecture diagram aur 'Future scope' zaroor daalo.
- Code GitHub pe rakho aur roz push karo.

Quick install (ek command mein)

```
pip install ultralytics supervision opencv-python easyocr fastapi uvicorn streamlit
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